

Project Proposal: Integrated IT Infrastructure Database

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1. Domain Description

For my semester project, I am designing a database to manage the IT infrastructure of a simulated company. As a student currently enrolled in Active Directory (ITN 113), Switching & Routing (ITN 155), and Ethical Hacking (ITN 261), I realized that my coursework generates three distinct types of data: user accounts, network topology, and security vulnerabilities. Currently, this data exists in separate silos—spreadsheets, Packet Tracer files, and VM logs.

My project, the **Integrated IT Infrastructure Database**, will consolidate this information into a single relational system. The goal is to track "who has access to what" and "which assets are vulnerable." Instead of generating random sample data, I plan to populate the tables with the actual configurations I am building in my other classes. For example, the **Users** table will be imported directly from my Windows Server lab work, and the **Subnets** will match my Cisco network designs.

The database will meet the Third Normal Form (3NF) standard by separating repeating groups like Operating System versions and Department names into their own lookup tables. This project will allow me to practice complex SQL queries—such as linking a specific user to the vulnerability status of their assigned workstation—creating a functional tool that mirrors the Asset Management systems used in real IT departments.

2. Potential Entities

1. **Hosts:** Physical hardware (Servers, Workstations) with specs like CPU/RAM.
2. **VirtualMachines:** The guest operating systems running on the hosts.
3. **OperatingSystems:** A reference table for OS names (e.g., Windows 11, Kali Linux).
4. **NetworkSegments:** Stores VLAN IDs and Subnet masks (from ITN 155).
5. **IPAllocations:** Links a VM's network interface to a specific IP address.
6. **Users:** Employee accounts and credentials (from ITN 113).
7. **Departments:** Organizational groups (Sales, IT, HR) for sorting users.
8. **Vulnerabilities:** A list of Common Vulnerabilities and Exposures (CVEs).
9. **ScanResults:** Records when a specific VM tests positive for a specific Vulnerability (from ITN 261).
10. **SoftwareLicenses:** Tracks expiration dates for installed software.

3. Preliminary Relationships

- **Hosts -> VirtualMachines (One-to-Many):** A physical server runs multiple VMs.
- **Departments -> Users (One-to-Many):** A department has many employees.

- **NetworkSegments -> IPAllocations (One-to-Many):** A subnet contains many IP addresses.
- **VirtualMachines <-> Vulnerabilities (Many-to-Many):** A VM can have many vulnerabilities, and a vulnerability can affect many VMs (resolved via the **ScanResults** bridge table).

4. Project Scope

Types of Users/Roles:

- **System Administrator (DBA):** Full access to create tables, modify schema, and import bulk data from lab exports.
- **Security Auditor (Read-Only):** Restricted access to execute **SELECT** queries for generating risk reports, but cannot modify or delete asset records.

Key Functionality:

- **Data Ingestion:** Importing user lists from CSV files generated in Active Directory.
- **Asset Tracking:** Querying which physical hosts are running specific virtual machines.
- **Risk Reporting:** Identifying outdated OS versions or unpatched vulnerabilities across specific subnets.

Sample Queries:

1. "Show me all users in the 'IT' department."
2. "List all Virtual Machines assigned to VLAN 10."
3. "Count how many machines have 'Critical' vulnerabilities."

5. Rationale

I selected this domain to connect my database studies with my current Networking and Security classes. Instead of creating fake data, I will populate the database iteratively throughout the semester using the actual configurations from my ITN 155 and ITN 261 labs. This approach allows me to build a realistic system that evolves with my coursework, helping me understand the complex relationships between users, infrastructure, and security risks.

6. Academic Integrity & AI Usage

I acknowledge the course academic integrity policies. **AI Disclosure:** I used Google Gemini to assist with brainstorming the initial list of entities and outlining the relationships between them. The final domain selection, project scope, and the strategy to integrate my specific coursework are my own original work.